We create chemistry

Safety data sheet

1. Identification

Product identifier

2EH Heavies Pasadena

Recommended use: industrial chemicals
Not recommended use: Not intended for sale to or use by the general public.

Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Operating Division Petrochemicals

Telephone: +49 621 60-42151
E-mail address: sds-petrochemicals@basf.com

Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

2. Hazards Identification

Classification of the substance or mixture

According to UN GHS criteria

Skin Corr./Irrit. 2

Eye Dam./Irrit. 1
Skin Sens. 1B
Repr. 2 (unborn child)
Aquatic Acute 3
Aquatic Chronic 3

For the classifications not written out in full in this section the full text can be found in section 16.

Label elements

Globally Harmonized System (GHS)

Pictogram:

Signal Word:
Danger

Hazard Statement:

H318	Causes serious eye damage.
H315	Causes skin irritation.
H317	May cause an allergic skin reaction.
H361	Suspected of damaging the unborn child.
H402	Harmful to aquatic life.
H412	Harmful to aquatic life with long lasting effects.

Precautionary Statements (Prevention):

P280	Wear protective gloves, protective clothing and eye protection or face protection.
P261	Avoid breathing dust/fume/gas/mist/vapours/spray.
P273	Avoid release to the environment.
P201	Obtain special instructions before use.
P202	Do not handle until all safety precautions have been read and understood.
P272	Contaminated work clothing should not be allowed out of the workplace.
P264	Wash contaminated body parts thoroughly after handling.

Precautionary Statements (Response):

P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P310	Immediately call a POISON CENTER or physician.
P303 + P352	IF ON SKIN (or hair): Wash with plenty of soap and water.
P362 + P364	Take off contaminated clothing and wash it before reuse.

Precautionary Statements (Storage):

P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

Labeling of special preparations (GHS):
The following percentage of the mixture consists of components(s) with unknown hazards regarding the acute toxicity: 31 - 37 %, oral

3. Composition/Information on Ingredients

Substances

Not applicable

Mixtures

Chemical nature

organic compounds

Hazardous ingredients (GHS)
According to UN GHS criteria

2-Ethylhexyl 2-ethylhexanoate

Content (W/W): >= 20 % - < 75 %	Repr. 2 (unborn child)
CAS Number: 7425-14-1	H361
EC-Number: 231-057-1	
INDEX-Number: 607-622-00-7	

Isobutyric acid, monoester with 2,2,4-trimethylpentane-1,3-diol

Content (W/W): > 0 % - < 25 %	Skin Irrit. 3
CAS Number: 25265-77-4	Aquatic Acute 3
EC-Number: 246-771-9	H316, H402

2-Ethylhexane-1,3-diol

Content (W/W): > 0 % - < 15 %	Acute Tox. 5 (oral)
CAS Number: 94-96-2	Eye Dam./Irrit. 1
EC-Number: 202-377-9	H318, H303
INDEX-Number: 603-087-00-9	

2-Ethylhexan-1-ol

Content (W/W): > 0 % - < 15 %	Flam. Liq. 4
CAS Number: 104-76-7	Acute Tox. 5 (oral)
EC-Number: 203-234-3	Acute Tox. 4 (Inhalation - mist)
	Skin Irrit. 2
	Eye Irrit. 2A
	STOT SE 3 (irr. to respiratory syst.)
	Aquatic Acute 3
	Aquatic Chronic 3
	H227, H319, H315, H332, H303, H335, H402, H412

2-Ethylhex-2-enal

Content (W/W): > 0 % - < 7 %	Flam. Liq. 3
CAS Number: 645-62-5	Acute Tox. 5 (oral)
EC-Number: 211-448-3	Skin Irrit. 2
	Skin Sens. 1B
	Aquatic Acute 3
	Aquatic Chronic 3
	H226, H315, H303, H317, H402, H412

For the classifications not written out in full in this section the full text can be found in section 16.

4. First-Aid Measures

Description of first aid measures

Remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention. Immediately administer a corticosteroid from a controlled/metered dose inhaler.

On skin contact:
Immediately wash thoroughly with plenty of water, apply sterile dressings, consult a skin specialist.

On contact with eyes:
Hold eyes open and rinse slowly and gently with water for 15 to 20 minutes. Remove contact lenses, if present, after first 5 minutes, then continue rinsing.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.

Information on: 2-Ethylhex-2-enal
Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.

Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:
foam, carbon dioxide, dry powder

Unsuitable extinguishing media for safety reasons:
water jet

Special hazards arising from the substance or mixture

Vapours may form explosive mixture with air. Cool endangered containers with water-spray.

Advice for fire-fighters

Special protective equipment:
Wear a self-contained breathing apparatus.

Further information:
Fire debris must be disposed of in accordance with offical regulations.

6. Accidental Release Measures

Personal precautions, protective equipment and emergency procedures

Handle in accordance with good industrial hygiene and safety practice. Avoid contact with the skin, eyes and clothing. Avoid inhalation.

Environmental precautions

Do not discharge into drains/surface waters/groundwater. Discharge into the environment must be avoided.

Methods and material for containment and cleaning up

For large amounts: Pump off product.
For residues: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr). Dispose of absorbed material in accordance with regulations.
Dispose of in accordance with national, state and local regulations.

7. Handling and Storage

Precautions for safe handling

Ensure thorough ventilation of stores and work areas.

Protection against fire and explosion:
Prevent electrostatic charge - sources of ignition should be kept well clear - fire extinguishers should be kept handy.

Conditions for safe storage, including any incompatibilities

Further information on storage conditions: Containers should be stored tightly sealed in a dry place.

8. Exposure Controls/Personal Protection

Control parameters

Components with occupational exposure limits

104-76-7: 2-Ethylhexan-1-ol
645-62-5: 2-Ethylhex-2-enal
7425-14-1: 2-Ethylhexyl 2-ethylhexanoate

Exposure controls

Personal protective equipment

Hand protection:
Wear chemical resistant protective gloves.

Eye protection:
Tightly fitting safety goggles (chemical goggles)., Wear face shield if splashing hazard exists.

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. head protection, apron, protective boots, chemical-protection suit.

General safety and hygiene measures
Avoid contact with the skin, eyes and clothing. Handle in accordance with good industrial hygiene and safety practice.

9. Physical and Chemical Properties

9.1. Information on basic physical and chemical properties

State of matter:	liquid	
Form:	liquid	
Colour:	pale yellow green, clear	
Odour:	characteristic	
Flash point:	98 °C	(ASTM D93, closed cup)
Solubility in water:	sparingly soluble	
Vapour pressure:	0,1 - 2 hPa	
Density:	0,9064 g/cm3 (20 °C)	

9.2. Other information

Information with regard to physical hazard classes

Corrosion to metals

Corrosive effects to metal are not anticipated.

10. Stability and Reactivity

Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals: Corrosive effects to metal are not anticipated.

Chemical stability

The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions

Violent reaction under influence of oxidizing agents.

Conditions to avoid

Avoid all sources of ignition: heat, sparks, open flame. Heating may cause pressure build-up and possible rupture of the container.

Incompatible materials

Substances to avoid:
strong oxidizing agents

Hazardous decomposition products

:
No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information

Information on toxicological effects

Acute toxicity

Information on: 2-Ethylhexan-1-ol
Assessment of acute toxicity:
Of low toxicity after single ingestion. Virtually nontoxic after a single skin contact. Of moderate toxicity after short-term inhalation.

The following percentage of the mixture consists of components(s) with unknown hazards regarding the acute toxicity: 31 - 37 %

Irritation

Assessment of irritating effects:
Causes serious eye damage. Causes skin irritation.

Information on: 2-Ethylhexane-1,3-diol
Assessment of irritating effects:
Not irritating to the skin. May cause severe damage to the eyes.

Information on: 2-Ethylhexan-1-ol
Assessment of irritating effects:
Eye contact causes irritation. Skin contact causes irritation.

Respiratory/Skin sensitization

Assessment of sensitization:
May cause allergic skin reaction.

Information on: 2-Ethylhex-2-enal
Assessment of sensitization:
Caused skin sensitization in animal studies.

Germ cell mutagenicity

Assessment of mutagenicity:
Based on available data, the classification criteria are not met.

Carcinogenicity

Assessment of carcinogenicity:
Based on available data, the classification criteria are not met.

Reproductive toxicity

Assessment of reproduction toxicity:
Based on available data, the classification criteria are not met.

Developmental toxicity

Assessment of teratogenicity:
May cause harm to the unborn child.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
No known chronic effects.

Other relevant toxicity information

The product has not been tested. The statements on toxicology have been derived from the properties of the individual components.

12. Ecological Information

Toxicity

Assessment of aquatic toxicity:
Harmful to aquatic life. Harmful to aquatic life with long lasting effects.

Information on: 2-Ethylhexan-1-ol
Assessment of aquatic toxicity:
Acutely harmful for aquatic organisms. Harmful to aquatic organisms based on long-term (chronic) toxicity study data. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Information on: 2-Ethylhex-2-enal
Assessment of aquatic toxicity:
Acutely harmful for aquatic organisms. The chronic aquatic risk classification is based on acute aquatic toxicity study data and the environmental fate properties of the product. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Information on: 1-Hexanol, 2-ethyl-, manuf. of, by-products from, distn. residues
Assessment of aquatic toxicity:
Acutely harmful for aquatic organisms. Inhibition of degradation activity in activated sludge is not to be anticipated during correct introduction of low concentrations.

Additional information

Other ecotoxicological advice:
The product has not been tested. The statements on ecotoxicology have been derived from the properties of the individual components.

13. Disposal Considerations

Waste treatment methods

Must be disposed of or incinerated in accordance with local regulations.

Contaminated packaging:
Uncleaned empties should be disposed of in the same manner as the contents.
Contaminated packaging should be emptied as far as possible; then it can be passed on for recycling after being thoroughly cleaned.
Uncontaminated packaging can be re-used.

14. Transport Information

Land transport

ADR

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

RID

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Inland waterway transport

ADN

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user:	None known

Transport in inland waterway vessel
Not evaluated

Sea transport

IMDG

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number or ID number	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

15. Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

Not applicable

16. Other Information

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

Skin Corr./Irrit.	Skin corrosion/irritation
Eye Dam./Irrit.	Serious eye damage/eye irritation
Skin Sens.	Skin sensitization
Repr.	Reproductive toxicity
Aquatic Acute	Hazardous to the aquatic environment - acute
Aquatic Chronic	Hazardous to the aquatic environment - chronic
Skin Irrit.	Skin irritation
Acute Tox.	Acute toxicity
Flam. Liq.	Flammable liquids
Eye Irrit.	Eye irritation
STOT SE	Specific target organ toxicity — single exposure
H361	Suspected of damaging the unborn child.
H316	Causes mild skin irritation.
H402	Harmful to aquatic life.
H318	Causes serious eye damage.
H303	May be harmful if swallowed.
H227	Combustible liquid.
H319	Causes serious eye irritation.
H315	Causes skin irritation.
H332	Harmful if inhaled.
H335	May cause respiratory irritation.
H412	Harmful to aquatic life with long lasting effects.
H226	Flammable liquid and vapour.
H317	May cause an allergic skin reaction.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.